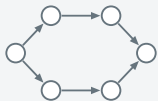


1. Rollout

π_t



Rollout + GAE

$\hat{A}_r, \hat{A}_{c_k}$

$\bar{J}_k$ observed episode cost

2. Flow inputs



Clustered states
Transition flow

$\hat{F}, A, C_k$

3. Constraints

Soft budgets

$$\langle C_k, F \rangle \leq b_k$$

$$b_k = B_k/L$$



$$F_H = 0$$

Hard exclusions

4. SafeOT projection

$$\min_{F \geq 0} -\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F})$$

s.t. flow balance + constraints



$\hat{F}$

F^*

Target flow is not executed

advantages

$$\bar{\lambda}_k^* \leftarrow \text{clip} + \text{EMA} \leftarrow \lambda_k^*$$

budget duals

5. Priced update

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+ \rightarrow \oplus \rightarrow \lambda_k \rightarrow \tilde{A} = \frac{\hat{A}_r - \sum_k \lambda_k \hat{A}_{c_k}}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO} \rightarrow \pi_{t+1}$$

next rollout